

TAHA ARAB

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SUMMARY

Senior Mechanical Engineering Honors student with hands-on experience taking hardware from concept through fabrication, integration, and test — including a plant-adopted mechanical fix designed during an industrial internship. U.S. citizen eligible for security clearance.

EDUCATION

University of Houston, Cullen College of Engineering – Houston, TX

Expected May 2027

B.S. Mechanical Engineering | Minor in Mathematics | GPA: 3.3 | Honors Program

Coursework: Finite Element Analysis, Thermodynamics, Fluid Mechanics, Solid Mechanics, Statics, Dynamics, Material Science & Lab, Experimental Methods & Lab, Numerical Analysis, Computing for Engineers (MATLAB)

WORK EXPERIENCE

INEOS Olefins & Polymers USA – Engineering Intern

May 2025 – Aug 2025

- Co-designed a thrust needle bearing assembly to replace plant-wide seized ball-and-socket joints on hotbox vents - identified corrosion and heat cycling as the root cause via root cause analysis (RCA), engineered a drop-in fix, and presented to management; adopted for new hotboxes
- Built Power Automate workflow automating contract migration and a SharePoint contractor database from scratch for 1,500+ records - cut vendor search time by 80%
- Negotiated rates with an environmental compliance vendor and conducted competitive bid analysis across two vendor pairs - reduced projected contractor spend by over 30%
- Developed MRO Vendor Dashboard in Excel via SAP data tracking top 30 vendors by annual spend; diagnosed and resolved a SAP character limit bug causing recurring data entry failures
- Shadowed maintenance PM on winches, compressor motor belt replacement, and bucket lift operations - hands-on exposure to industrial mechanical systems across the facility

PROJECTS

CAD-Based Plane Fabrication

Jan 2025 – May 2025

- Led all SolidWorks modeling and assembly integration for a 4-person team - fabricated components via 3D printing, CNC, and laser cutting within strict weight and clearance constraints; earned maximum score
- Managed project timeline and task delegation across a 4-person team; identified and resolved 3 tolerance/fit conflicts during assembly by adjusting part dimensions and re-fabricating components, contributing to a max-score result

Automated Sorting Machine (Design-Build-Test)

Jan 2025 – May 2025

- Designed and built a fully functional remote-triggered electromechanical system - integrated servos, LEDs, and Arduino microcontroller from scratch
- Diagnosed a non-functional system mid-test by methodically testing components part-by-part with a multimeter, isolating the faulty connection and restoring full functionality under time pressure

OUTPOST - Browser-Based Tactical FPS (outpostfps.vercel.app)

Aug 2026 – Present

- Built a 3D tactical shooter from scratch in JavaScript using Three.js for rendering and the Rapier physics engine for character collision and hitscan raycasting; implemented headshot hitbox detection, weapon recoil, and skeletal character animation (idle/run/shoot/death) with crossfade blending
- Engineered three tiers of enemy bot AI (Easy/Medium/Hard) with independent aim-spread, reaction time, and cover-seeking logic; procedurally generated arenas that scale layout to match size (1v1/3v3/5v5) and iteratively shipped the project across 10 tracked development milestones

Personal Engineering Portfolio (tahaarab.vercel.app)

Aug 2026 – Present

- Designed and built a personal portfolio site (Next.js, React, Tailwind CSS) featuring an interactive AI chatbot "digital twin" that answers recruiter questions about my background and technical projects

SKILLS

Design & CAD: SolidWorks, Fusion 360, SketchUp, OpenSCAD - assemblies, tolerancing (GD&T), detailed drawings

Fabrication: 3D Printing (Ultimaker Cura), CNC Machining, Laser Cutting (RDWorks)

Embedded & Electronics: Arduino programming, servo/LED integration, remote-triggered actuation, circuit wiring and debugging

Laboratory: Tensile, fatigue, Charpy impact, creep, corrosion, heat treatment; pressure transducer & thermocouple calibration; velocity distribution in ducts; stress/strain measurement

Analysis & Computing: MATLAB, Tableau, SQL, R, Excel, SAP, Power Automate, JavaScript, React, Next.js

EXTRACURRICULAR & COMMUNITY

Young Muslims - NeighborNet Coordinator | Core Team

Jan 2022 – Present

- Organized weekly events, retreats, and camps for 50-200 youth; collaborated with regional leadership on city-wide planning

Community Speaker

Aug 2023 – Present

- Monthly sermons delivered to congregations of up to 800 people; YouTube channel with 15,000+ views

HONORS & AWARDS

JP Morgan Merit Scholarship (Aug 2026) | UH Foundation Excellence Scholarship (Jan 2026) | Cizik Scholarship (Aug 2025) | Google Data Analytics Certificate (Aug 2024) | Dean's List (Jan 2024)